

An unbiased comparison of immunoglobulin sequence aligners

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Abstract

Adaptive Immune Receptor Repertoire sequencing (AIRR-seq) is critical for our understanding of the adaptive immune system's dynamics in health and disease. Reliable analysis of AIRR-seq data depends on accurate rearranged immunoglobulin (Ig) sequence alignment. Various Ig sequence aligners exist, but there is no unified benchmarking standard representing the complexities of AIRR-seq data, obscuring objective comparisons of aligners across tasks. Here, we introduce GenAIRR, a modular simulation framework for generating Ig sequences alongside their ground truths. GenAIRR realistically simulates the intricacies of V(D)J recombination, somatic hypermutation, and an array of sequence corruptions. We comprehensively assessed prominent Ig sequence aligners across various metrics, unveiling unique performance characteristics for each aligner. The GenAIRR-produced datasets, combined with the proposed rigorous evaluation criteria, establish a solid basis for unbiased benchmarking of immunogenetics computational tools. It sets up the ground for further improving the crucial task of Ig sequence alignment, ultimately enhancing our understanding of adaptive immunity.

Keywords: immunoglobulin; sequence alignment; AIRR-seq; V(D)J recombination; benchmarking; somatic hypermutation

Introduction

The adaptive immune system functionality relies upon a diverse and dynamic set of cell receptors. In lymphocytes, this diversity originates from the V(D)J recombination process [1], with B cells undergoing further diversification through affinity maturation; a process that includes clonal expansion [2], somatic hypermutation (SHM) [3], and affinity-dependent selection [4]. Advances in sequencing technologies, particularly adaptive immune receptor repertoire sequencing (AIRR-seq) [5], have profoundly enhanced our understanding of this repertoire, providing detailed insights into its dynamics and diversity in response to a wide spectrum of immunological challenges [6–13].

Analysing AIRR-seq data requires an accurate alignment of rearranged immunoglobulin (Ig) sequences to their germline allele ancestors. This task poses significant computational challenges due to factors such as the stochastic nature of gene trimming during V(D)J recombination [14], alterations introduced by SHM [15], ambiguities resulting from sequencing errors [16], and the vast array of known germline allele sequences, as well as the existence of yet to be discovered allele sequences, which further complicate the alignment process [17].

To address these challenges, two primary approaches are utilized for aligning Ig sequences: string distance metrics-based and Hidden Markov Models (HMM)-based. Distance-based methods [18–20], while computationally efficient, may encounter difficulties with complex sequence variations such as insertions,

deletions, and mutations. In contrast, HMM-based methods [21–23] leverage probabilistic models to capture some of the stochastic nature of V(D)J recombination and sequence evolution. This approach provides a more detailed representation of the real diversity of Ig sequences and somatic evolutionary patterns. However, these methods typically require more computational resources and rely on parameters inferred from empirical, potentially noisy, datasets.

All tools for rearranged Ig sequence alignment require a germline allele reference set that encompasses the known alleles expected to be included in AIRR-seq data. This germline reference set is used to establish the metrics necessary for the alignment. Current germline reference sets from the International ImMunoGeneTics Information System (IMGT) [24] and the Open Germline Receptor Database (OGRDB) [25] are fundamental resources for immunogenetics research, which differ in their curation methodologies. IMGT is a comprehensive database that provides detailed information on the structure, function, and classification of immunoglobulins, T cell receptors, and major histocompatibility complex (MHC) molecules, with a strong emphasis on historical data. The IMGT Ig allele reference set is curated by the IMGT team. The allele reference sets available through OGRDB on the other hand are curated by the inferred allele review committee (IARC) of the AIRR community. Despite significant efforts, both of these resources still do not have a fully comprehensive or nearly complete allele set for humans,

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let alone for other species of interest [26]. Thus, an adaptable germline reference set is essential for several reasons. First, numerous more recent studies have identified novel Ig alleles that are not present in standard reference databases (e.g. [27–32]). These newly identified alleles significantly contribute to immune repertoire diversity and play a crucial role in accurate alignment and analysis in personalized genomics [23, 33–35]. Recent reviews [36] have discussed these findings in detail, emphasizing their importance in the context of immunogenetics. The importance of personalized genomics cannot be overstated, as individuals may have unique variations in their immunoglobulin genes, affecting immune responses and disease susceptibilities [37–43]. Furthermore, the ability to modify the reference to accommodate personalized genotypes ensures precise alignment and interpretation of Ig sequences. This adaptability also assists in identifying rare and low-frequency variants that may be critical to immune function but are often disregarded in standard reference-based alignments during immune repertoire analysis [44].

Understanding whether a rearranged Ig sequence is productive, or expressed, is vital for various aspects of immunological research. Since many factors contribute to the ability to express an Ig protein, only experimental validation can confirm the productivity status of a sequence. Nevertheless, several necessary conditions must be met for a sequence to be considered productive, which were identified originally by IMGT. The standardized framework to computationally infer the productivity of Ig sequences [45] includes ensuring a correct open reading frame; the absence of aberrations in the start codon, splicing sites, and regulatory elements; the absence of internal stop codons; and an in-frame junction region where the V, D, and J gene segments align properly. Despite these standardized criteria, variations in the assessment of sequence productivity can arise due to differences in algorithms and methodologies used by the different sequence aligners. Factors such as the handling of ambiguous gene segment boundaries, treatment of sequencing errors, and interpretation of junction regions can lead to discrepancies in sequence productivity classifications among aligners.

To address these challenges and accurately evaluate alignment tools, past benchmarks often used datasets derived from or simulated based on direct sequencing efforts [19, 23, 46]. These datasets inherently carry biases, like unequal allele representation and batch-effects prevalent in many cohorts. In addition, these benchmarks often overlooked critical aspects such as the ability of aligners to handle insertion or deletions (indels), accurately define the start and end positions of segments, and stratify performance across different levels of SHM. Further, the IGH/IGK/IGL genomic loci display high variability among individuals, posing challenges in creating representative reference sets. Such challenges underscore the need for a more comprehensive and unbiased benchmarking approach. The benefits of using data generated through controlled and reproducible simulations in such tasks are of paramount importance [47, 48].

This is not to say that the commonly seen distributions in real sequencing data are not valuable for evaluating aligners under typical scenarios. In fact, they can be highly useful for assessing performance in more common biological contexts. However, when the goal is to eliminate confounding factors that may arise from allele or noise distributions specific to a given dataset, a uniform distribution allows for a more rigorous evaluation. It provides a clearer picture of an aligner's ability to detect all alleles in a given reference, highlighting which alleles pose a challenge and which can be more reliably detected.

Building upon these insights, we propose a two-fold approach to address the challenges in benchmarking alignment tools effectively. First, we present a benchmarking setup that encompasses three critical metrics for evaluating sequence alignment tools: 1) Assessment of the tools' accuracy in correctly identifying sequence allele calls. This precision is fundamental, as it forms the basis for understanding the alleles and genes at play, with significant downstream impacts on analyses such as genotype determination [35, 49], haplotype inference [50], cloning [16, 51], and SHM calls. 2) Segmentation: aiming to precisely identify the start and end of alleles within the sequence. Precision in this task is crucial because incorrect segmentation, such as premature or late trimming of the 3' end of the V allele can lead to missed SHM events, influence the productivity assessment, or erroneous identification of non-real SHM events. 3) Productivity assessment of sequences. Downstream analysis pipelines commonly filter out what they consider to be non-productive sequences, hence correct assessment by the aligners has a high impact. Although evaluating productivity may seem straightforward, differences between aligners arise from variations in their algorithms and implementations. Second, we introduce GenAIRR, a robust simulation framework designed to generate Ig sequence datasets with established ground truths that enable accurate and comprehensive comparisons among aligners. GenAIRR incorporates realistic sequence corruptions and noise, filling gaps in existing simulation frameworks [23, 52–57] and providing a solid foundation for a robust benchmarking setup. See [Supplementary Table 1](#) for an overview of existing simulation frameworks.

This manuscript aims not only to elevate the standards of aligner comparison, but also to establish a comprehensive framework for the ongoing evaluation of both existing and newly developed alignment methodologies. By doing so, it seeks to significantly improve the precision and reliability of AIRR-seq analysis. Such advancements will deepen our understanding of the adaptive immune system's responses to pathogens and enhance our ability to leverage this knowledge in health and disease contexts.

Methods

Simulation workflow of GenAIRR

The GenAIRR simulation framework consists of a series of customizable modular events designed to generate rearranged Ig sequences that reflect the complexity and variability found in natural immune responses. Each event is controlled by a set of adjustable parameters, providing a high degree of flexibility and coverage.

The simulation begins with a stochastic selection of variable (V), diversity (D), and joining (J) alleles from a representative pool. In the context of the current manuscript, the aim was to generate datasets with a uniform representation of all alleles. Once selected, the allele sequences undergo specific trimming processes to enhance their structural complexity. Additionally, the nontemplated, palindromic (NP) regions are integrated using Markov chains derived from empirical data, which model the probabilistic arrangement of nucleotides (see [Supplementary Section 1](#)). Next, the simulation incorporates controlled mutation rates, models of SHM, random substitutions, and indels to enrich sequence diversity. GenAIRR also provides the option to introduce ambiguous 'N' bases deliberately, to test the robustness

Table 1. Illustration of the main GenAIRR sequence simulation workflow. Each numbered step corresponds to a stage in Fig. 1. Steps marked as 'Optional' can be adjusted or omitted based on specific simulation needs, and steps marked under 'Productivity Test' indicate whether a productivity assessment is performed at this step

Step	Process	Optional	Productivity Test
1-2	Generate Naive Sequence	X	✓
1-2	Adjust Segment Positions Post-Recombination	X	X
1-2	Include Indistinguishable V, J alleles from trimming	X	X
1-2	Include Indistinguishable D alleles from 5' and 3' trimming	X	X
1-2	Flag Sequence with Short D Alleles	X	X
3-4	Simulate SHM	✓	✓
3-4	Simulate Sequencing/Library Prep Errors	✓	✓
5	Introduce Insertions/Deletions	✓	✓
6-7	Apply Corruption to 5' End of V Allele	✓	✓
8	Introduce Ambiguous 'N' Bases	✓	X
8	Calculate Finalized Sequence Mutation Rate	X	X

of sequence alignment algorithms. Additionally, sequencing artifacts are simulated, particularly at the 5' ends of V alleles to mimic shorter sequences and 5' untranslated region residuals. These features are introduced through tunable parameters that control the type, probability, and extent of errors. A detailed overview of the simulation steps is provided in Table 1, and an example of the tunable parameters used in this manuscript is provided in Supplementary Table 2. It's important to note that thanks to GenAIRR's modularity, as more accurate noise models for specific sequencing platforms, or SHM models, become available, they can be easily integrated into GenAIRR. This modularity allows researchers to further refine the simulation and tailor it to specific experimental conditions, improving the overall robustness and applicability of the simulated data.

Ground truths produced by GenAIRR

GenAIRR generates detailed ground truth data for each Ig sequence, which is essential for evaluating the accuracy of sequence alignment tools. These data include information on the V, D, and J allele calls, and their respective segment start and end positions. In addition, the ground truth includes documentation of the mutations, trimmings, indels, productivity, and corruption events (see Supplementary Table 4 for ground truth example).

During the simulation of Ig sequences, GenAIRR incorporates several strategies to address challenges such as the potential ambiguities arising from trimming or corruption at the allele ends. One significant challenge is the short length of D alleles, caused by their substantial trimming at both the 5' and 3' ends during recombination. This often results in sequences retaining only a minimal number of bases, making it impossible to distinguish these short sequences from multiple D alleles. GenAIRR manages this by employing a hyperparameter threshold to determine the minimal length for a D allele to be considered distinguishable post-recombination. Alleles shorter than this threshold are labeled as 'Short-D'. The chosen threshold of five bases reflects a balance between computational efficiency and biological realism, covering approximately 85% of D alleles, as seen in Supplementary Figure 1. A map object post-trimming identifies which alleles become indistinguishable under specific scenarios, and adjustments to the ground truth are made. This includes corrections to the start and end positions of each allele following recombination and the simulated insertion of NP regions to verify that the trimmed section was not partially reconstructed by the NP regions. Such careful adjustments maintain the integrity of

the ground truth, enabling more precise and reliable simulation outcomes (see Supplementary Section 2.1).

In addition to addressing these challenges, GenAIRR assesses the productivity of the simulated sequence. The criteria for a productive sequence are the absence of stop codons, the sequence to be in the correct open reading frame, and the presence of two conserved amino acids (AA) before and after the Complementary Determining Region 3 (CDR3). The region encompassing these conserved AAs and CDR3 is termed the junction. Sequence productivity is assessed four times during the GenAIRR sequence simulation (refer to Table 1, and Supplementary Fig. 2).

Benchmarking setup

For creating our benchmarking setup, we have generated three datasets, created four comparison matrices, and selected suitable alignment tools for comparison.

Data preparation

Using GenAIRR, two datasets, each containing 6 million sequences, were simulated for this study. These datasets were generated using the AIRR-C reference set [17] and clustered using the Allele Similarity Cluster method [44] to remove identical sequences. Within this reference set, there are 192 V alleles, 33 D alleles, and 7 J alleles. The selection of alleles for the simulated sequences followed a uniform distribution. This approach aimed to achieve an equal representation of all alleles in the reference and to eliminate any potential biases.

The first dataset, DS1, comprised purely productive sequences, reflecting common AIRR-seq data, enabling us to evaluate alignment tools under typical conditions. Supplementary Table 2 shows the parameters used to generate this dataset. The second dataset, DS2, primarily consisted of non-productive sequences resulting from recombination, SHM events, or noise introduction (Supplementary Table 2). This allowed us to test the alignment tools under extreme conditions and assess their performance. The third dataset, DS3, was used to assess productivity calls in a combination of data from DS1 and DS2. The fourth dataset, DS4, was generated to evaluate the aligners with sequences simulated using empirical parameters derived from real-world datasets. In summary:

- **DS1:** Includes only productive sequences without introduced corruptions, representing optimal alignment scenarios.
- **DS2:** Contains sequences with intentional corruptions like ambiguous base maskings ('N's), 5' corruption events

(removal or addition of nucleotides), and indels. This dataset comprises mostly non-productive sequences, with a minor fraction of productive sequences.

- **DS3:** A combined dataset comprising sampled sequences from both DS1 and DS2. Specifically, four million productive sequences were randomly sampled from DS1, along with four million non-productive sequences from DS2.
- **DS4:** Includes only productive sequences without corruptions, generated using empirical parameters (IGHV gene usage, 5' and 3' gene trims, and mutation rates) derived from real-world datasets. See [Supplementary section 2.3](#).

Benchmarking immunoglobulin sequence alignment tools

In this study, we surveyed the existing alignment tools ([Supplementary Table 3](#)), and selected four popular tools for comparison. These tools were selected not only because of their popularity, but also because they are consistently maintained and developed in the immunogenetics community, and also comply with the AIRR community schema for annotated AIRR-seq data [58]. The leading aligners that were evaluated here are IgBLAST[19], MiXCR [20], HighV-QUEST [18] by IMGT, and Partis [23].

Benchmarking criteria

To benchmark the performance of the aligners, we have used four metrics:

- **Alignment Accuracy:** The effectiveness with which each aligner identified V, D, and J alleles under varying conditions of mutation rates and sequence corruptions was evaluated. Accuracy was quantified using an 'Agreement' metric, defined for each sequence as 1 if the intersection of ground truth alleles (G) and alleles predicted by the aligner (P) is not empty $G \cap P \neq \emptyset$, and 0 otherwise. This metric directly measures the ability of an aligner to correctly identify alleles used to generate the sequence. (This is due to the assumptions that when there are multiple alleles in the ground truth, they are all ambiguous to one another)
- **Segmentation Accuracy:** We evaluated aligners' precision in segmenting gene regions, focusing on the accuracy of segment start and end positions amidst mutations and sequence corruptions. This assessment is restricted to sequences for which there was a perfect match between the aligners' allele calls for V, D, and J and the ground truth. In cases where the aligner made multiple assignments for any of V, D, or J alleles, the sequence was included in the comparison only if the first assignment matched the ground truth. This approach was adopted because aligners' segmentation values are associated with the first assignment; hence, including sequences with erroneous values from subsequent assignments would skew the analysis. An adjustment was made for Partis due to intentional N-padding at the beginning of sequences. The results were modified to align Partis' segmentation positions accurately with the ground truth by subtracting the added N-padding. To quantify segmentation errors, we utilized the Root Mean Square Error (RMSE) metric, defined as:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (P_i - G_i)^2}$$

where n is the number of sequences, P_i is the predicted segment position by an aligner, and G_i is the actual ground truth position.

- **Productive Sequence Detection Accuracy:** This evaluation involved the calculation of the proportion of true positives and true negatives, which were then utilized to determine the overall classification accuracy. The results are presented using confusion matrices.
- **Runtime Efficiency and Resource Utilization:** The time needed to process 6 million sequences was measured for each aligner to compare computational efficiency. This evaluation was conducted on our cluster with specifications including an Intel(R) Xeon(R) Gold 6130 CPU @ 2.10GHz, 64 CPU cores, and 376 GB of RAM. The aligners were run on a single thread without parallelization, to align 60,000 sequences. The processing time for this subset was then multiplied by 100 to estimate the runtime for 6 million sequences.

Results

Creating a benchmarking setup using GenAIRR

To establish a robust benchmarking setup, a bias-free dataset is required. For this, we created GenAIRR, an AIRR-seq data simulator that simulates the full spectrum of V(D)J recombination events and introduces realistic sequence corruptions such as 5' nucleotide trimming or addition, masking nucleotides with Ns, and introduction of indels. GenAIRR mitigates biases by enabling simulations without relying on empirical data distribution, opting instead for a uniform distribution to generate sequences (refer to [Supplementary Table 1](#) for comparisons with other simulation tools). In addition, GenAIRR's modular architecture allows users to tailor simulations to reflect specific experimental conditions. GenAIRR provides comprehensive ground truth data for each simulated sequence, including allele calls, segmentation positions, and productivity assessments, formatted in the AIRR community schema for annotated AIRR-seq data [58]. This facilitates straightforward comparisons with the output of commonly used alignment tools. GenAIRR is illustrated in [Figure 1](#).

Using GenAIRR, we created two datasets, each containing 6 million sequences, with a uniform distribution of alleles. The first dataset (DS1) consisted solely of productive sequences, devoid of corruptions, N masks, or indels, but did include varying mutation rates to mimic real AIRR-seq data. The second dataset (DS2) included mainly nonproductive sequences, resulting from the rearrangement process, introduction of mutations, or corruption events such as 5' trimming or addition, N insertions, and indels (refer to [Supplementary Table 2](#) for simulation details).

We used the GenAIRR report feature to check how alleles are distributed in these datasets. Although we aimed for an even distribution, we noticed a small difference in the usage of certain V and J alleles in DS1 ([Fig. 2A](#) and [C](#)) compared to DS2 ([Supplementary Fig. 3A](#) and [C](#)). This difference stems from constraints inherent in the generation process of productive sequences. In addition, out of the 198 V alleles in the OGRDB reference set, eight V alleles could not produce productive rearrangement due to a stop codon within the allele sequence.

In simulating D alleles, the protocol involved trimming both the 5' and 3' ends, sometimes resulting in very short sequences that pose alignment challenges due to their potential to match multiple alleles (see method section 2.2). In actual AIRR-seq data, it is impossible to ascertain the origin of these short D sequences. Hence, GenAIRR incorporates a feature that identifies sequences of five or fewer nucleotides as 'Short-D' and conceals their origin. In both DS1 ([Fig. 2B](#)) and DS2 ([Supplementary Fig. 3B](#)), the allele usage across other D alleles was generally uniform. The simulated datasets exhibited a distribution of CDR3 lengths that resembled empirical data ([Fig. 2D](#), [Supplementary Fig. 3D](#)).

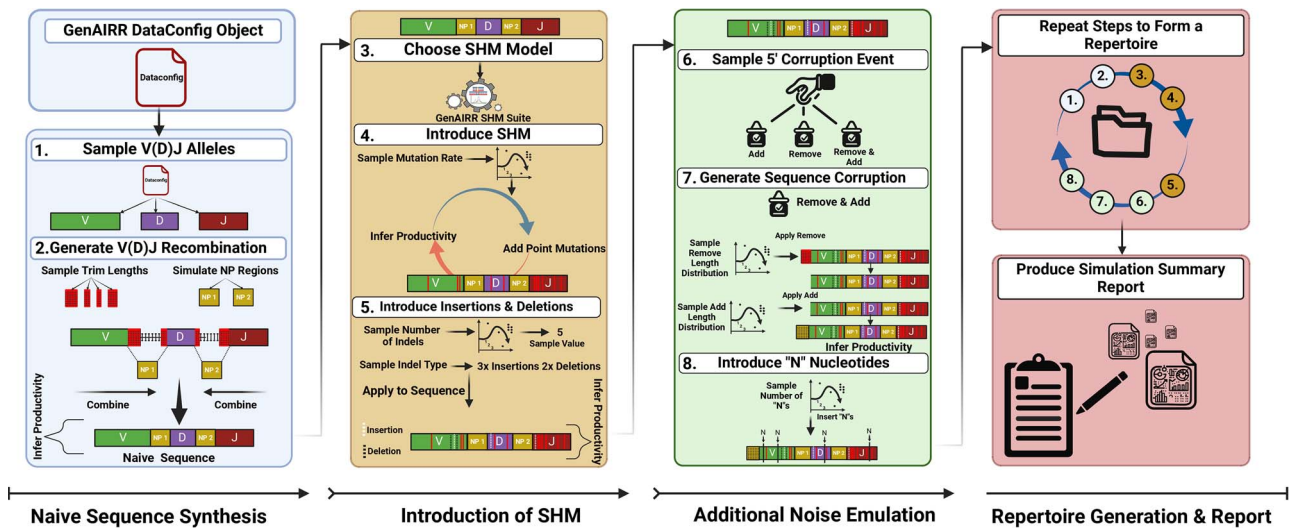


Figure 1. GenAIRR modular architecture to simulate rearranged Ig sequences. The first column (steps 1–2) describes simulation of a naive Ig sequence using the provided configuration file (DataConfig). The second column (steps 3–5) illustrates the introduction of alterations to the simulated sequence, such as SHM and indels. The third column (steps 6–8) illustrates the introduction of further experimental noise to the simulated sequence, including 5' corruptions and N nucleotides. Finally, the fourth column illustrates that GenAIRR allows for repeating the sequence simulation to form a repertoire and generating a report summarizing its statistical properties (Created with BioRender.com).

Benchmarking immunoglobulin sequence aligners

In the presented benchmark evaluation, we first surveyed the existing alignment tools, and selected four widely used aligners, IgBLAST [19], MiXCR [20], HighV-QUEST [18], and Partis [23], based on their popularity, compatibility with the AIRR community schema for annotated AIRR-seq data [58], consistent support, and active development within the immunogenetics community (see [Supplementary Table 3](#) for a summary). Importantly, these aligners were also chosen for their diverse alignment methodologies. IgBLAST employs BLAST for alignment, HighV-QUEST and MiXCR utilize multiple sequence alignment techniques, and Partis adopts an HMM approach. Here, we utilized the newly published reference set by the AIRR community [17], which requires the aligners to enable the use of a custom reference set. HighV-QUEST is the only aligner of the four that is unable to accept a custom reference set, and thus in cases where the resulting allele assignment HighV-QUEST produced was not included in the AIRR-C reference set, we matched it with the closest allele in the set. Not all aligners return results for every sequence in the datasets. In DS1, all aligners achieved a perfect retrieval rate at the V allele level, successfully providing assignments for all 6 million sequences (Fig. 2E). Furthermore, Partis consistently had a retrieval rate of 100% for both the D and the J alleles, closely followed by HighV-QUEST with ~ 99%. The retrieval rate of IgBLAST for J alleles is roughly 99% and 96% for the D alleles. MiXCR showed a lower retrieval rate at 92% for J alleles and 91% for D alleles (Fig. 2F–G). The trend remained the same for DS2, with Partis returning the highest retrieval rate, trailed by IgBLAST, then IMGT, and lastly MiXCR ([Supplementary Fig. 3E–G](#)).

Precision of aligners varies with segment mutation rates and lengths

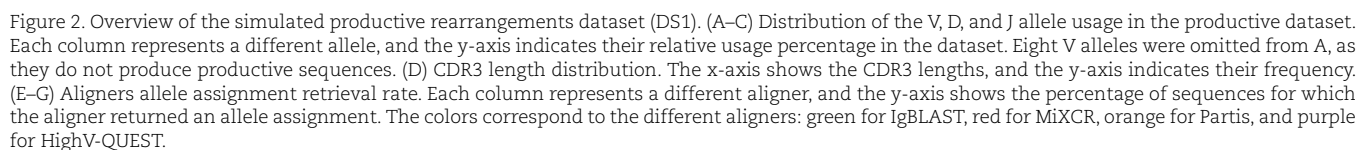
We evaluated the aligners' ability to accurately assign V, D, and J allele calls for each sequence from simulated productive rearrangements dataset (DS1). We observed variations in performance across different mutation rates and sequence lengths (Fig. 3). IgBLAST demonstrated the highest overall accuracy in assigning

V alleles. Notably, HighV-QUEST exhibited underperformance at lower mutation rates but showed a relatively steady, good accuracy in highly mutated sequences (Fig. 3A).

Assigning the J allele was a relatively easier task due to the reduced variability among alleles. Partis consistently outperformed all aligners across various mutation rates, with IgBLAST closely following but showing a decline in sequences with over a 17% mutation rate. MiXCR maintained good accuracy up to a mutation rate of 11%, but, as with V alleles, its performance declined at higher mutation rates compared to other aligners (Fig. 3B).

The evaluation of D assignment performance is shown as a function of D segment length rather than mutation rate. This is because mutation rates can vary significantly due to the short segment lengths, potentially masking the influence of both factors (Fig. 3C). For longer D sequences, Partis and IgBLAST exhibited slightly superior performance, with MiXCR lagging behind. For shorter D segments, IgBLAST remained the top performer, while HighV-QUEST exhibited the lowest agreement with ground truth. These findings underscore the nuanced performance dynamics of aligners depending on mutation rates and sequence lengths. To further assess the impact of segment length on accuracy, we isolated sequences that did not encounter any mutations within theIGHD segment (Fig. 3D). The accuracy of all aligners spiked across all segment lengths, with IgBLAST demonstrating accuracy above 90% for sequences longer than 9 nucleotides. However, as observed in Fig. 3C, accuracy declines in short D segments, emphasizing the influence of segment length on the ability to correctly assign the D allele. Aligner's precision in DS2 yielded similar results ([Supplementary Figure 4](#)). Differences in all three allele segment classifications can be partially attributed to the fact that IgBLAST and HighV-QUEST return more than one call in cases where there are ties in their alignment scores, unlike Partis and MiXCR, which typically return only one call by default ([Supplementary Figure 15](#)).

To test how the uniform allele distribution and the wide range of mutation rates impact assignment accuracy in typical data sets, we generated a new simulated dataset (DS4, [Supplementary Section 2.3](#)) based on empirical metrics derived from actual



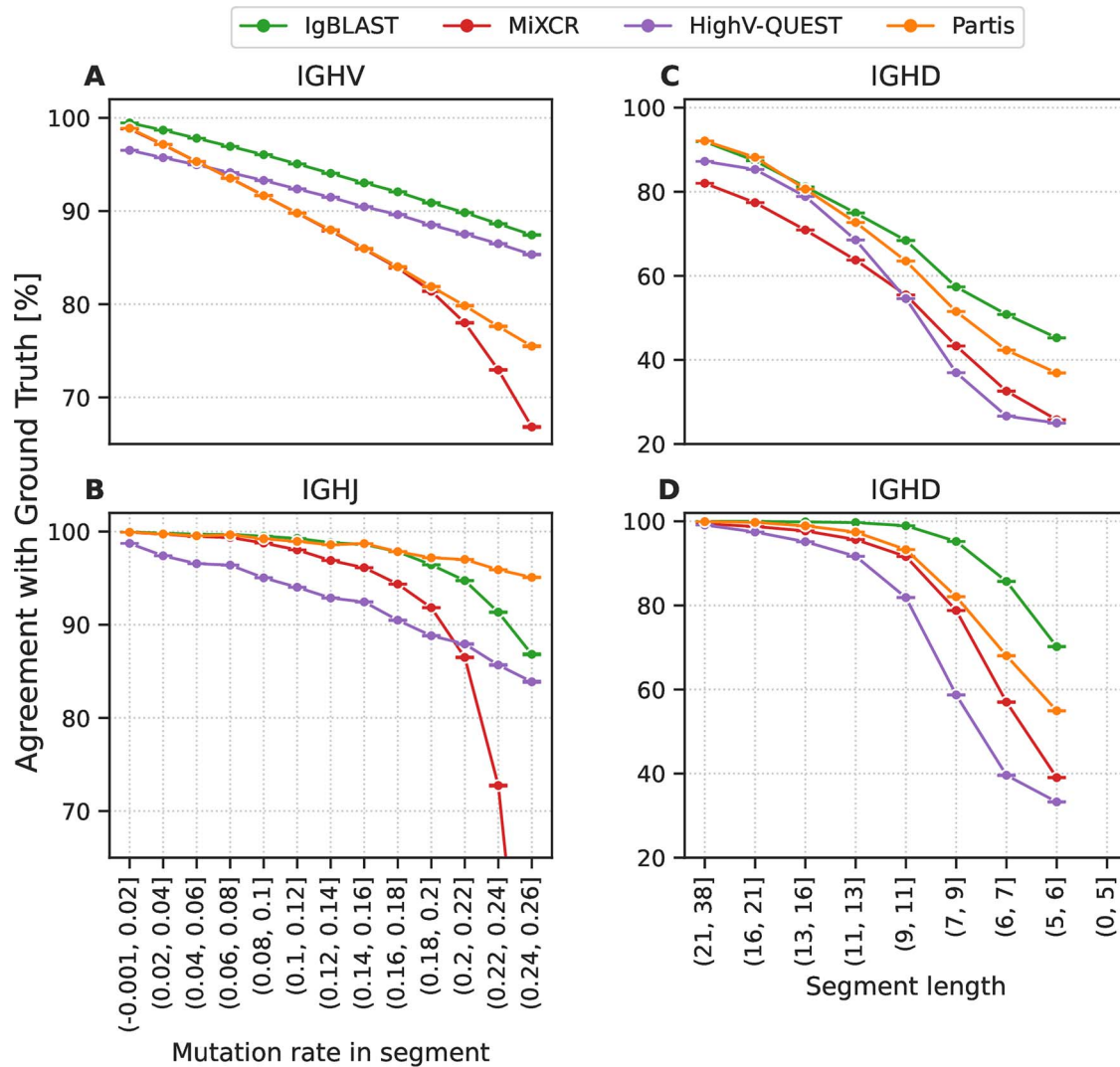


Figure 3. Aligners predictive performance of allele calls. (A+B) Agreement with ground truth for IGHV (A) and IGHJ (B) allele segment calls across varying mutation rates. The x-axis displays tuples of values, where each tuple (X, Y) represents the borders of an x-axis bin. The values within these bins correspond to mutation rates, and the y-axis shows the agreement percentage with the true allele call. The colors correspond to the aligners as indicated in the figure. (C) Agreement with ground truth for IGHD allele segment calls across varying segment lengths. The x-axis represents the varying D segment lengths from 38 to 6, and the y-axis shows the agreement percentage with the true allele call. (D) Same as C) for sequences with 0 mutations in their D segments.

data. These simulations, which mimic real-world allele usage and mutation rates, provided valuable insights into how the aligners perform under more biologically realistic conditions. We used DS4 to evaluate the assignment accuracy of the four aligners (Supplementary Figures 6). The results showed that, aside from a slight improvement in D accuracy for Partis, bringing it closer to IgBlast, the overall ranking of accuracy across the aligners remained consistent. This suggests that the accuracy results are not primarily driven by the V gene distribution or mutation rate. However, this does not exclude the possibility of other contributing factors, which may still influence aligner performance in specific data sets.

In addition to evaluating overall Agreement, we further analysed the confusion among allele calls in DS1 to understand the specific challenges faced by each aligner at varying mutation rates (Supplementary Figures 11, 12, 13, 14). IGHJ4*02 and IGHJ5*02, for example, were consistently confused with each other and with IGHJ1*01 across all aligners. IgBLAST demonstrated superior performance, with lower confusion rates at low to mid mutation levels. On the other hand, MiXCR and Partis

showed higher confusion rates with these alleles but exhibited less variance at higher mutation rates. V alleles such as IGHV1-8*03, IGHV1-69*17, IGHV1-24*01, IGHV4-4*01, IGHV4-61*01, as well as other alleles in IGHVF6 and IGHVF3 clusters (IMGT Alleles in Supplementary Table 5), presented significant challenges, with even minor mutations, leading to substantial misassignments. Again, IgBLAST and Partis generally performed better, minimizing confusion, though all aligners struggled with these closely related allele groups (Supplementary Figures 7, 8, 9, 10). This detailed confusion analysis highlights the varying degrees of robustness among the aligners, particularly in distinguishing between highly similar alleles under different noise levels.

Segmentation position accuracy differs among aligners

Besides accurately identifying alleles in AIRR-seq alignment, another crucial task is segmenting the sequence into distinct allele regions, which facilitates the correct attribution of SHM events. To evaluate the accuracy of segmentation by the aligners, we utilized both DS1 and DS2. We computed the Root Mean

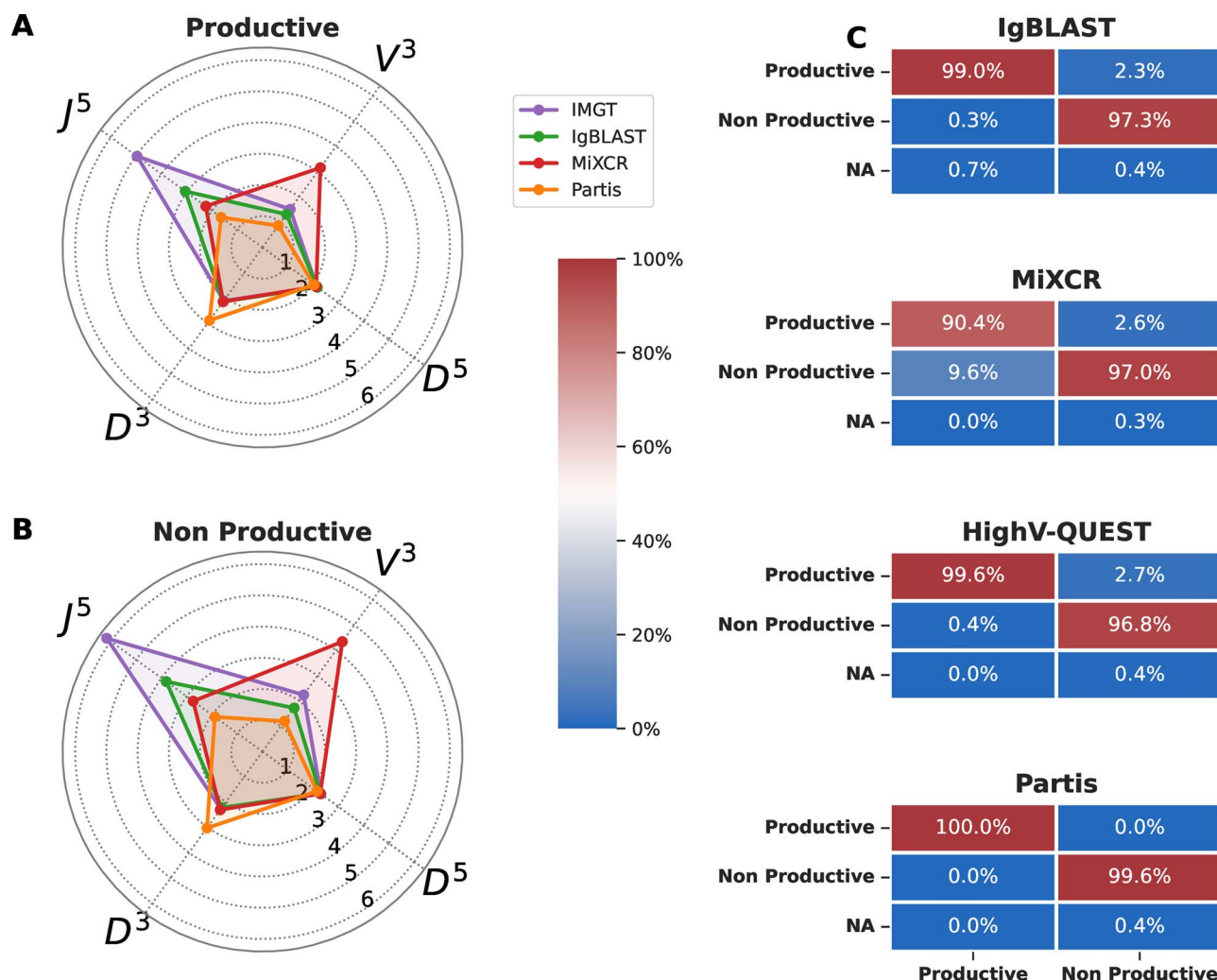


Figure 4. Aligners segmentation and productivity assessment. (A+B) Radar charts illustration of the RMSE (Root Mean Square Error) values for segmentation accuracy across gene segment start and end positions (V3, D3, D5, J3) in productive (A) and non-productive (B) sequences. (C) The confusion matrices provide an accuracy comparison of productive assessments. In each matrix, columns represent the 4M sequence subset of the simulated dataset, while rows represent the aligners' productive assessment categorized as Productive, Non Productive, or NA (indicating no assessment for the sequence). The value in each cell indicates the percentage of agreement with the ground truth. The color scale reflects the level of accuracy.

Square Error (RMSE) for each segment's 5' and 3' positions across all aligners. To ensure the accuracy and reliability of the analysis, we focused exclusively on sequences where allele calls perfectly matched the ground truth, prioritizing the first matching call in cases of multiple allele calls and excluding V 5' and J 3' segmentations from this analysis.

In DS1, the RMSE results for the V 3' position were generally similar across the aligners, except MiXCR, which exhibited a higher value (Fig. 4A). The accuracy in segmenting the D 5' position was consistent across all aligners. However, for the D 3' position, Partis showed a slightly higher RMSE value compared to the others. Regarding the J 5' position, HighV-QUEST recorded the highest RMSE, while Partis demonstrated the lowest value. In DS2, the aligners maintained relatively similar RMSE and ranking values, as observed in DS1 (Fig. 4B). To complement this analysis, we also calculated the accuracy of each aligner in predicting the exact start and end positions of each gene (Supplementary Figure 17). When focusing on accuracy rather than average RMSE, we observe that IMGT and IgBLAST perform similarly in productive sequences, while MiXCR and Partis exhibit distinct characteristics. Notably, Partis, despite having the lowest average RMSE for J'5,

tends to make predictions that are close but do not precisely align with the ground truth. A similar trend is seen for the D'5 and D'3 positions. In contrast, MiXCR shows a significant difference only in V'3 predictions. These patterns become more pronounced in non-productive sequences due to the additional sequence corruption events. IgBLAST, on the other hand, demonstrated the most consistency, maintaining stable accuracy across both productive and non-productive datasets. These findings underscore the significant disparities in segmentation accuracy across aligners and their performance across different dataset contexts.

Aligners accuracy in determining sequence productivity

Sequence productivity is important for immune repertoire analysis tools, affecting their precision in tasks such as genotype, haplotype, clonality, and diversity assessments. To assess the accuracy of the aligners in evaluating sequence productivity, we initially established the ground truth using GenAIRR. GenAIRR determines sequence productivity based on the following criteria: absence of

Table 2. Comparison of different IG sequence aligners focusing on criteria such as custom reference acceptance, operational efficiency, and accuracy across mutation rates (MR). Accuracy metrics are presented for productive (left) and non-productive (right) sequences, with the best results highlighted in **bold**. Root Mean Square Error (RMSE) quantifies the precision of segment start and end position predictions; lower values indicate higher precision. The runtime parameters were generated using a computer with an Intel(R) Xeon(R) Gold 6130 CPU @ 2.10GHz, 64 CPU cores, and 376 GB of RAM. The aligners were run on a single thread without parallelization

Criteria	IgBLAST		MiXCR		HighV-QUEST		partis	
	Prod.	Non	Prod.	Non	Prod.	Non	Prod.	Non
Accepts Custom Reference	✓		✓		✗		✓	
6M Seqs Runtime (Minutes)	2200		200		–		5400	
V Accuracy (MR < 10%)	97.79	97.57	95.29	93.84	94.93	94.61	95.31	93.43
V Accuracy (MR > 10%)	90.24	89.38	77.09	75.11	87.96	87.67	80.77	79.83
J Accuracy (MR < 10%)	99.74	99.61	99.45	99.41	96.83	95.14	99.62	99.24
J Accuracy (MR > 10%)	93.69	92.49	73.53	72.21	88.48	86.58	97.0	96.86
D Accuracy (Length < 10)	51.14	49.42	33.88	31.33	29.53	26.73	43.57	42.47
D Accuracy (Length > 10)	85.3	84.16	69.90	66.19	80.36	75.52	84.69	83.81
V 3' RMSE	1.31	1.72	3.15	4.35	1.51	2.24	0.86	1.2
D 5' RMSE	2.17	2.29	2.12	2.28	2.15	2.31	2.03	2.16
D 3' RMSE	2.14	2.22	2.14	2.3	2.11	2.27	2.89	3.02
J 5' RMSE	3.06	3.81	2.24	2.74	4.96	6.16	1.64	1.89
Productivity Accuracy	99	97.3	90.4	97	99.6	96.8	100	99.6

a stop codon, presence of conserved AA at the start and end of the junction, and alignment that is in-frame (Supplementary Figure 2). To assess aligner accuracy in determining productivity, we used dataset DS3, which contains four million randomly sampled productive sequence from DS1, and a matching number of randomly sampled nonproductive sequence from DS2.

For productive sequences, Partis demonstrated perfect accuracy, whereas IgBLAST showed a low error rate with a slightly higher rate of missing sequences, marginally outperforming HighV-QUEST, which had a higher error rate but no missing sequences. Lastly, MiXCR exhibited a higher error rate (Fig. 4C, left column). MiXCR higher error rate mostly stems from a lack of J assignment in the sequences that results in a non-productive assessment for the sequences. The pattern was comparable for nonproductive sequences. Partis showed the highest accuracy, with a small proportion of sequences missing. IgBLAST performed slightly better than HighV-QUEST and MiXCR in terms of error rates (Fig. 4C right column).

Discussion

The current study represents an unbiased evaluation of Ig sequence aligners, facilitated by the GenAIRR simulation framework. This comprehensive assessment across multiple metrics has outlined distinct strengths and weaknesses inherent in the widely used aligners IgBLAST, MiXCR, HighV-QUEST, and Partis. The datasets generated by GenAIRR provide a robust, unbiased platform for benchmarking these aligners in a spectrum of challenging immunogenetic variables, mimicking real-world complexities. This benchmarking analysis explores how noise levels, such as mutation rates and corruption events, affect aligner agreement with the ground truth.

GenAIRR's design enables the simulation of repertoires with a uniform allele distribution, providing a balanced assessment of an aligner's performance across all alleles. While real sequencing data often exhibit a skewed distribution, which biases aligner evaluation towards more frequently observed alleles, a uniform distribution ensures even the rarest alleles are thoroughly tested. This feature is particularly important for users seeking

confidence in the aligner's ability to detect alleles that are rare but still biologically relevant. That said, it is important to note that the benchmarking simulations incorporated empirically derived parameters for trimming of gene segments. Simulations using empirically derived allele usage profiles resulted in consistent results with those observed under uniform conditions, reinforcing that uniform simulations do not distort the overall trends but provide an alternative means of testing an aligner's ability to detect rare alleles.

Another important advantage of using a uniform allele distribution is that it allows us to identify the true sources of aligners' confusion between alleles. As detailed in the results, we present confusion matrices (Supplementary Section 7) to highlight key differences between the aligners, under varying mutation rates. By using a uniform allele distribution, we can infer the most common sources of confusion across all alleles without bias, whereas using real data would have introduced bias toward the more frequently occurring alleles.

The aligners' comparisons (Table 2) revealed varying performances. MiXCR emerged as the most efficient aligner in terms of runtime, however IgBLAST outperformed the other aligners in almost all allele assignment accuracies. One exception is Partis, which demonstrated excellent performance in J assignment accuracy across both low and high mutation rates. The accuracy metric used here for allele assignment is relatively lenient, requiring a single assignment per sequence to match the ground truth. This leniency can disadvantage aligners that, by default, return only one assignment, such as Partis and MiXCR, as they have a lower probability of matching the ground truth. Note that erroneous assignments can, in many cases, be rectified by inferring a personal genotype.

Properly segmenting a sequence can influence factors such as attributing SHM events to the correct segments and assessing sequence productivity. Partis showed the lowest RMSE for position predictions in most categories, and the highest accuracy in the productivity assessment. Error rates in productivity can be attributed not only to segmentation errors, which can lead to missed stop codons, but also to misclassifying sequences that lack conserved residues in the junction as productive.

GenAIRR's modularity is a standout feature, allowing the simulation to start from a naive sequence and progressively introduce noise (Table 1). This modular design facilitates easy adaptation of the code to different noise models and addition of optional stages, such as simulating epigenetic modifications, incorporating complex recombination patterns, or adding sophisticated SHM models. GenAIRR is released as an open-source code to encourage community contributions to the simulation framework, potentially enhancing its versatility and utility beyond the application presented here. The parameters used to generate the data for our analyses are detailed in Supplementary Table 2, and the code is available on GitHub. We encourage the community to utilize these resources to simulate datasets and benchmark future versions of their aligners, both existing and future ones.

Although a comparison was made between GenAIRR and other simulators in terms of runtime (Supplementary Figure 16), a more comprehensive and systematic comparison of all possible features, performance, and assumptions of various simulators lies beyond the scope of this work and should be conducted by independent third parties. Such comparisons are more effective when carried out impartially, rather than by developers of a specific tool.

Here, we used GenAIRR to simulate human Ig sequences, but other species can present different challenges. For example, macaque monkeys have many more known V(D)J alleles [59] compared to human, and as such can affect aligner performances in a non-trivial manner.

In conclusion, our study not only presents a robust benchmarking setup for refining existing alignment tools and encouraging the development of new ones, but also underscores the immense potential of GenAIRR's modular and adaptable framework. By integrating advanced functionalities and addressing key challenges in simulating Ig sequences, we pave the way for more accurate and reliable bioinformatics tools in immunogenetics. This collaborative effort, coupled with standardized benchmarking criteria using simulated sequences with known ground truth, propels the field towards optimized algorithms and deeper insights into immune system analysis, ultimately benefiting healthcare and disease research.

Key Points

- We introduce GenAIRR, a modular simulation framework for Ig sequence analysis that generates adaptive immune receptor sequences with realistic V(D)J recombination, somatic hypermutation, and sequence corruption.
- Using GenAIRR, we generated datasets of noisy immunoglobulin (Ig) heavy chain sequences alongside their respective ground truths in terms of the parameters used to simulate each of them.
- Publicly available Ig aligners were reviewed, and the leading four in terms of usability and support were chosen for a cross-comparison on the generated datasets.
- Three criteria to test aligner performance were defined. Namely, correctly identifying sequence allele calls, segmentation, and productivity assessment of sequences.
- Our comprehensive assessment of the prominent Ig sequence aligners across various metrics, unveiled unique performance characteristics for each aligner.

Supplementary data

Supplementary data is available at *Briefings in Bioinformatics* online.

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Code availability

The GenAIRR package is open-source and available for Python users. It can be installed via pip or accessed directly from GitHub at <https://github.com/MuteJester/GenAIRR>.

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